

Advanced MS Excel Functions & Data Analysis (Exam + Job Oriented)

1. IFERROR

Handles errors gracefully.

=IFERROR(A1/B1,"Error")

2. SUMIF

Conditional sum.

=SUMIF(A1:A10,">50",B1:B10)

3. SUMIFS

Multiple conditions sum.

=SUMIFS(C1:C10,A1:A10,"Sales",B1:B10,">1000")

4. COUNTIF

Conditional count.

=COUNTIF(A1:A10,"Pass")

5. COUNTIFS

Multiple conditions count.

=COUNTIFS(A1:A10,"Pass",B1:B10,">40")

6. TEXT

Formats numbers as text.

=TEXT(A1,"dd-mm-yyyy")

7. TEXTJOIN

Joins text with delimiter.

=TEXTJOIN(", ",TRUE,A1:A5)

8. UNIQUE

Extracts unique values.

=UNIQUE(A1:A20)

9. FILTER

Filters data dynamically.

=FILTER(A1:C20,B1:B20>50)

10. SORT

Sorts data dynamically.

=SORT(A1:C20,2,-1)

11. SORTBY

Sort by another column.

=SORTBY(A1:C20,C1:C20,-1)

12. SEQUENCE

Generates sequence of numbers.

=SEQUENCE(10,1,1,1)

13. LET

Improves formula readability.

=LET(x,A1+B1,x*10)

14. LAMBDA

Creates custom function.

=LAMBDA(x,y,x*y)(5,6)

15. OFFSET

Returns dynamic range.

=SUM(OFFSET(A1,0,0,5,1))

16. INDIRECT

Creates reference from text.

=SUM(INDIRECT("A1:A10"))

17. CHOOSE

Returns value by index.

=CHOOSE(2,"Red","Blue","Green")

18. TRANSPOSE

Switches rows and columns.

=TRANSPOSE(A1:C3)

19. POWER QUERY (Data Tool)

Used for data cleaning, merging, transformation (ETL).

20. PIVOT TABLE (Data Tool)

Used for summarizing large datasets efficiently.

Bonus: What Examiners & Interviewers Love

- XLOOKUP vs VLOOKUP
- INDEX + MATCH
- Dynamic Arrays (FILTER, UNIQUE, SORT)
- Error handling using IFERROR
- Data analysis using Pivot Tables